

MOSES THIONG'O

Geospatial Science | Earth Observation | Data Analytics | Applied Research | AI Training & Evaluation | Decision Support
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English (Fluent), Swahili (Fluent), Gikuyu (Native)

PROFESSIONAL SUMMARY

A multidisciplinary technical professional with experience spanning food systems, geospatial science, Earth observation, data analytics, applied research, AI evaluation and development-focused decision support. Combines advanced spatial and quantitative methods with domain knowledge across agriculture, natural resources, climate resilience, infrastructure and environmental systems to translate complex evidence into practical planning and investment decisions. Experienced in designing reproducible analytical workflows; integrating geospatial, Earth observation, environmental, climate, disaster-risk, agricultural and socio-economic datasets; developing decision-support products; and communicating technical findings through maps, dashboards, research and technical documentation. Brings together analytical depth, systems thinking and hands-on technical execution across research, programme and technology environments.

CORE PROFESSIONAL EXPERTISE

GIS Platforms: ArcGIS Pro · ArcGIS Enterprise · ArcGIS Online · ArcMap · QGIS · Survey123 · ArcGIS Experience Builder · ArcGIS Dashboards

Spatial Data Quality: Attribute validation · Geometry validation · Coordinate systems · Projections · Spatial reference troubleshooting · Completeness review · Consistency review · Duplicate detection · Anomaly detection · Data correction

Geodatabases & Data Management: ArcGIS geodatabases · Enterprise geodatabases · PostgreSQL · PostGIS · SQL · Data loading · Data interoperability · Metadata standards · Version control

Mapping & Spatial Analysis: Vector analysis · Raster analysis · Geoprocessing · Remote sensing · Land-use analysis · Land-cover analysis · GPS field mapping · Map editing · Thematic cartography · Cartographic layouts · Dashboards

Programming & Automation: Python · ArcPy · GeoPandas · Rasterio · R · SQL · Batch scripting · Reproducible GIS workflows

Data Analytics & Visualization: Power BI · Tableau · ArcGIS Dashboards · Data visualization · Analytical reporting · Decision-support outputs

Earth Observation & Environmental Analysis: Google Earth Engine · Sentinel-2 · Copernicus · OpenEO · WEKEO · ClimateSERV · Change detection · Environmental monitoring

Technical Delivery: Quality assurance · Technical documentation · Workflow design · Knowledge-base writing · Training · Stakeholder communication · Remote collaboration · Cross-functional delivery

PROFESSIONAL EXPERIENCE

QGIS Expert and AI Trainer

Micro1 Inc. | Remote | April 2026 - July 2026

- Led end-to-end GIS projects covering spatial data acquisition, cleaning, validation, coordinate systems, geoprocessing, symbology, cartographic layout production, and final-output quality.
- Reviewed source datasets and geospatial outputs against technical requirements, identifying attribute, geometry, CRS, classification, completeness, and consistency issues.
- Validated maps and datasets through source-to-output reconciliation, feature and extent checks, coordinate-system verification, output inspection, and reproducible quality-assurance procedures.
- Created workflow guides, QA checklists, reviewer notes, and troubleshooting records that made technical decisions traceable and repeatable.
- Managed multiple technical deliverables independently while collaborating with distributed project teams.
- Performed human validation of AI-generated technical tasks across specialist application environments, extending evaluation work beyond GIS into Microsoft 365, LibrePCB and Qucs-S, and assessing outputs for correctness, completeness, reproducibility and adherence to application-specific requirements.
- Developed, wrote and validated end-to-end deterministic technical tasks and evaluation rubrics for Microsoft 365, LibrePCB and Qucs-S against detailed quality and execution requirements.

GIS Specialist, THRIVE 2030 Kenya Project

World Vision International | Nairobi, Kenya | June 2024 - January 2026

- Worked closely with the Design, Monitoring, Evaluation, Accountability and Learning team on programme monitoring, donor reporting, national planning, integrated data infrastructure, and evidence-based decision support.
- Designed and managed GIS infrastructure combining household, environmental, land-use, and spatial datasets for more than 300,000 households.
- Managed version control and change tracking across large spatial datasets and reporting documents, strengthening consistency, traceability, and quality across programme cycles.
- Validated GPS and field records, investigated location and attribute inconsistencies, and produced corrected datasets, maps, dashboards, and analytical outputs.

- Trained more than 70 field staff and developed step-by-step guidance to improve field data collection, validation, documentation, and geolocation accuracy.
- Designed and deployed an ArcGIS Experience Builder application with embedded dashboards to support near-real-time visualization and programme decision-making.
- Produced thematic maps, analytical reports, visualizations, and spatial evidence for donor reporting and stakeholder engagement.
- Co-produced national vulnerability and risk maps integrating environmental indicators, disaster-risk data and household-level exposure information to support FY26–FY30 planning and geographic prioritization.

Urban Planner and GIS Analyst

United Nations Human Settlements Programme (UN-Habitat) | Nairobi, Kenya | May 2023 - November 2023

- Reviewed, integrated, and validated multi-source topographic, land-cover, socioeconomic, road, boundary, and coastal-planning datasets.
- Applied metadata, classification, and data-quality standards across planning datasets.
- Produced thematic maps supporting Local Physical and Land Use Development Plans and evidence-based urban planning.
- Developed and maintained a Spatial Data Hub and ArcGIS Dashboard for stakeholder engagement, data sharing, programme reporting, and evidence-based planning.
- Designed GIS database structures and organized spatial datasets for efficient retrieval, reporting, and stakeholder delivery.
- Produced maps, spatial outputs, workflow documentation, and technical briefs for cross-functional urban, coastal, and infrastructure-planning teams.

Spatial Data Analyst

REGID Carbon Limited | Nairobi, Kenya | March 2023 - May 2023

- Analyzed, classified, and validated georeferenced data and satellite-derived outputs for ecosystem-restoration and carbon-accounting projects.
- Applied Earth Observation and change-detection techniques to assess deforestation, land-use transitions, and environmental change.
- Supported work across Kenya, Zimbabwe, and Senegal while maintaining accuracy, traceability, and documentation standards.

GIS Analyst

Environmental Systems Research Institute (Esri) | Redlands, California, USA | February 2020 - February 2023

- Supported global users of ArcGIS Pro, ArcGIS Enterprise, ArcMap, and ArcGIS Online across infrastructure, land-management, environmental-monitoring, and related sectors.
- Reviewed technical cases, user-provided datasets, geodatabases, diagnostic information, and workflow descriptions to reproduce issues and isolate root causes.
- Diagnosed and resolved spatial-reference conflicts, attribute inconsistencies, schema problems, data-integrity errors, and enterprise-geodatabase workflow issues.
- Resolved a substantial share of weekly GIS support cases while contributing to standardized troubleshooting and technical-support workflows.
- Documented defects, reproduction steps, severity, corrective actions, and validation evidence for product and engineering teams.
- Led and contributed to cartographic quality-assurance efforts supporting consistent, high-quality map outputs.
- Authored technical guides and knowledge-base articles while delivering work independently across distributed time zones.

SELECTED RESEARCH & PROFESSIONAL EXPERTISE

Spatial Strategies for Peace

- Applied spatial analysis to armed-group activity, cross-border movement, and displacement patterns in urban and peri-urban areas of Eastern Democratic Republic of the Congo through research with the Harvard Humanitarian Initiative.

Geospatial Research & Applied Remote Sensing

- Research experience spans NOAA ROV dive-track modelling, Fall Armyworm spread mapping, land-use change monitoring, and satellite-based flood classification with ArcGIS Pro and Python.

AI Training, Evaluation & Quality Assurance

- Experience evaluating AI-generated outputs against detailed rubrics and technical standards, producing structured feedback, documenting quality issues, developing prompts and workflows, and supporting consistent model-training and evaluation processes.

EDUCATION

MSc, Geographic Information Science (GIS) and Cartography December 2019

University of Redlands | Redlands, California, USA

Thesis: *Spatial Representation of NOAA's Remotely Operated Vehicles (ROVs) Dive Tracks*

Advisors: Mark Kumler, Ph.D. · Douglas Flewelling, Ph.D.

Award: *Jack Dangermond GIS Scholarship*

Bachelor of Environmental Planning and Management December 2018

Kenyatta University | Nairobi, Kenya

Research: *Mapping the 2017 spreading pattern of Fall Armyworm (*Spodoptera frugiperda*) and its implications on maize in Molo, Nakuru County*

Project Supervisor: Prof. Simon Mang'erere Onywere, Ph.D.

Award: *Esri 2018 GIS Young Scholar Award*

SELECTED CERTIFICATIONS AND PROFESSIONAL TRAINING

- **Esri** | ArcGIS Pro Associate 2101 (June 2022)
- **Anthropic** | Claude Code 101 (August 2026)
- **SurveyCTO Academy** | Foundations of SurveyCTO (April 2026)
- **FAO** | Global Forest Resources Assessment 2025 (March 2026)
- **McKinsey & Company** | McKinsey.org Forward Program (December 2025)
- **The World Bank Group** | Documenting Development Data Using Metadata Standards (December 2025)
- **Cisco** | Introduction to Modern AI (August 2025)
- **Google** | Foundations of Project Management (December 2022)